THE LECA RESURRECTION PROTOCOL

Recovery of the Last Eukaryotic Common Ancestor from Living Developmental Programs — One-Page Summary

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Every eukaryotic organism carries the Last Eukaryotic Common Ancestor (~1.5 Ga) as the base state of its developmental program. Placing any eukaryote in conditions geometrically derived from that attractor — before the experiment runs — stabilises it there. This is deliberate de-extinction of the oldest organism on Earth from a pre-specified geometric prediction, confirmed by a $200–400 experiment executable at any BSL-1 community biology laboratory in days. Pre-registered 2026-03-12.

The Theorem

The developmental trajectory from zygote to adult IS the evolutionary trajectory of the lineage, compressed into developmental time (the Reproduction Theorem). Therefore:

- **E** — **Evolutionary grade** — the attractor the lineage occupied at that point in its history.
- **D** — **Developmental position** — the equivalent stage in the developmental program today.

$E \equiv D$ (compressed). Arresting at position D stabilises grade E. The LECA sits at D=0: before the first multicellular step, in every eukaryote alive. **Conditions derived before the experiment. Prediction locked before the experiment. Result confirms the geometry.**

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The Three Pre-Registered DOIs

Document	DOI
Pre-Registration v1.0	...18986790
Registered Amendment A1	...18989573
The Ribosome Is the LUCA	...18988844

Repository: github.com/Eric-Robert-Lawson/ attractor-oncology

The Five Pre-Registered Predictions

- **P1** Arrest of any eukaryote before multicellular organisation produces a single-cell, nucleated, viable LECA-grade organism. Arm A alone.
- **P2** Arrested organisms from animal, plant, and fungal starting material are *morphologically indistinguishable* under light microscopy.
- **P3** *Genomically distinct* by sequencing — same attractor, different substrates.
- **P4** LECA-grade conserved genes active; all lineage-specific post-LECA genes silenced.
- **P5** Blind panel cannot identify starting kingdom from morphology alone (above 1/3 chance).

What Arm A Alone Establishes

Arm A is not a dormancy experiment. The arrest medium is derived geometrically from the LECA attractor before the experiment: Precambrian ionic composition, no nitrogen (germination program cannot fire), microaerophilic (<3% O_2, pre-GOE), 20–22 °C.

A spore in YPD at 30 °C is *waiting to become a fungus.* A spore in LECA-proxy conditions is *stabilised as the LECA* — downstream programs suppressed, only the conserved 4,000–6,000 gene eukaryotic base program running.

Arm A confirmed = first deliberate, pre-specified recovery of the LECA as an intentional target. Oldest de-extinction in history. $225 M has not produced a living mammoth (extinct 4,000 yr). Arm A recovers a 1.5 Ga organism for $200–400 in days — because the oldest target is the easiest to recover.

The Three Experimental Arms

Arm	Start	Cost	Timeline
A — Fungal	*S. cerevisiae*	$200–350	24–72 h
B — Animal	Mouse zygote	$400–600	24–72 h
C — Plant	*Arabidopsis*	$200–400	48–72 h

Arm A: P1 confirmed. De-extinction proof of concept. BSL-1. No credentials required. Arms A+B+C: morphologically identical across kingdoms, genomically distinct. Definitive proof.

Arm A — What You Actually Do

Day 0–3: Sporulate *S. cerevisiae* in 1% KAc, 25 °C, 48–72 h. Confirm asci by microscopy.

Day 3: Wash into arrest medium: 10 mM NaCl, 5 mM KCl, 2 mM $MgCl_2$, 1 mM $CaCl_2$, 0.1 mM $FeSO_4$, 1 mM KH_2PO_4, 0.1% glucose, **no nitrogen**, pH 6.5–7.0, 20–22 °C, <3% O_2.

Days 4–10: Observe at 24 h, 48 h, 72 h. Phase contrast (40×) + DAPI. Pass: ≥80% single-cell, nucleus confirmed, ≤10% germination, viable = **P1 confirmed.**

Key control: At 72 h add nitrogen to half the culture — germination resumes, proving stabilisation not cell death.

Equipment: Microscope, incubator, laminar flow. All at ChiTownBio, Chicago.

The geometry came first. The prediction was locked before the experiment. The LECA was never lost. It was running. This is the first time anyone looked deliberately.